PSEUDO-COHERENCE DESCENDS ALONG FAITHFULLY FLAT RING MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Finiteness of modules descends along faithfully flat ring maps. Algebra, Lemma \ref{algebra-lemma-descend-properties-modules}, cited at proof lines 16469-16470. Turns finiteness of the base changed cohomology module into finiteness over the ring.
- The m-pseudo-coherent complexes are those with vanishing cohomology above m and finitely generated cohomology in degree m. Lemma \ref{lemma-recognize-pseudo-coherent}, cited at proof lines 16463-16464. Gives the base case of the induction.
- Finiteness of cohomology in the top nonvanishing degree characterises pseudo-coherence one degree at a time. Lemma \ref{lemma-finite-cohomology}, cited at proof lines 16468-16469. Provides finiteness of the top cohomology module after base change.
- Cone lemma for pseudo-coherence in a distinguished triangle. Lemma \ref{lemma-cone-pseudo-coherent}, cited at proof lines 16477 and 16482-16483. Keeps the base changed cone m-pseudo-coherent and finishes the induction.

Source reference. Original lemma 068R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8015–8015. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 064W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7880–7880. Navigation pointer to a statement the

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7876–7876. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 03C4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4371–4371. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7875–7875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of R-modules. Let $R \to R'$ be a faithfully flat ring map. If the complex $K^\bullet \otimes_R R'$ is m-pseudo-coherent (resp. pseudo-coherent), then $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent).*

Proof. We will use without further mention that $- \otimes_R R'$ is an exact functor and that therefore

$$H^i(K^\bullet) \otimes_R R' = H^i(K^\bullet \otimes_R R').$$

Assume $K^\bullet \otimes_R R'$ is m-pseudo-coherent. Let $n \in \mathbf{Z}$ be the largest integer such that $H^n(K^\bullet)$ is nonzero; then n is also the largest integer such that $H^n(K^\bullet \otimes_R R')$ is nonzero. We will prove the lemma by induction on $n - m$. If $n < m$, then the lemma is true by Lemma 064W. If $n \geq m$, then $H^n(K^\bullet) \otimes_R R'$ is a finite R'-module, see Lemma 064S. Hence $H^n(K^\bullet)$ is a finite R-module, see Algebra, Lemma 03C4. Choose a finite free R-module E and a surjection $E \to H^n(K^\bullet)$. As E is projective we can lift this to a map of complexes $\alpha : E[-n] \to K^\bullet$. Then the cone $C(\alpha)^\bullet$ has vanishing cohomology in degrees $\geq n$. On the other hand, the complex $C(\alpha)^\bullet \otimes_R R'$ is m-pseudo-coherent, see Lemma 064R. Hence by induction we see that $C(\alpha)^\bullet$ is m-pseudo-coherent as a complex of R-modules. Applying Lemma 064R once more we conclude. □